Phyloseq Visualizations in Galaxy

BIOL 4810

2026-04-16

Goal: Use **phyloseq in Galaxy** to visualize your microbial community data with **alpha diversity**, **beta diversity**, and a **taxonomic bar plot**.
Submission: Fill in answers, upload screenshots where prompted, then click **Prepare submission preview** and **Download as PDF (Print)**.

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0) Your dataset

0A. Sample set / project name:

WCSMicrobiome

0B. Grouping variable used for comparisons:

SampleSource

0C. Other metadata variable(s) available for plotting or faceting:

BirdID, Date, InfectionStatus, HL, WBC, ParasiteLoad, Polychromasia, Sex, Control, Status, Year, Dominant, Sub

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phyloseq workflow focus

For this module, you will generate and interpret:

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Alpha diversity → Beta diversity (NMDS using Bray–Curtis) → Taxonomic
bar plot
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Reminder: These visualizations help you describe diversity within samples, differences among samples, and which taxa are driving those patterns.

Part 1 — Alpha diversity

1A) Alpha diversity

In your own words, what is alpha diversity?

The diversity of organisms in a given area/sample/spot.

1B) Observed richness

What does Observed measure?

Observed richness is a measure of the number of different species in a sample or site.

1C) Shannon diversity

What does Shannon measure?

Shannon Diversity measures how diverse a community is by considering both species richness and the number of members for each species. This measure assigns greater weight to rare species.

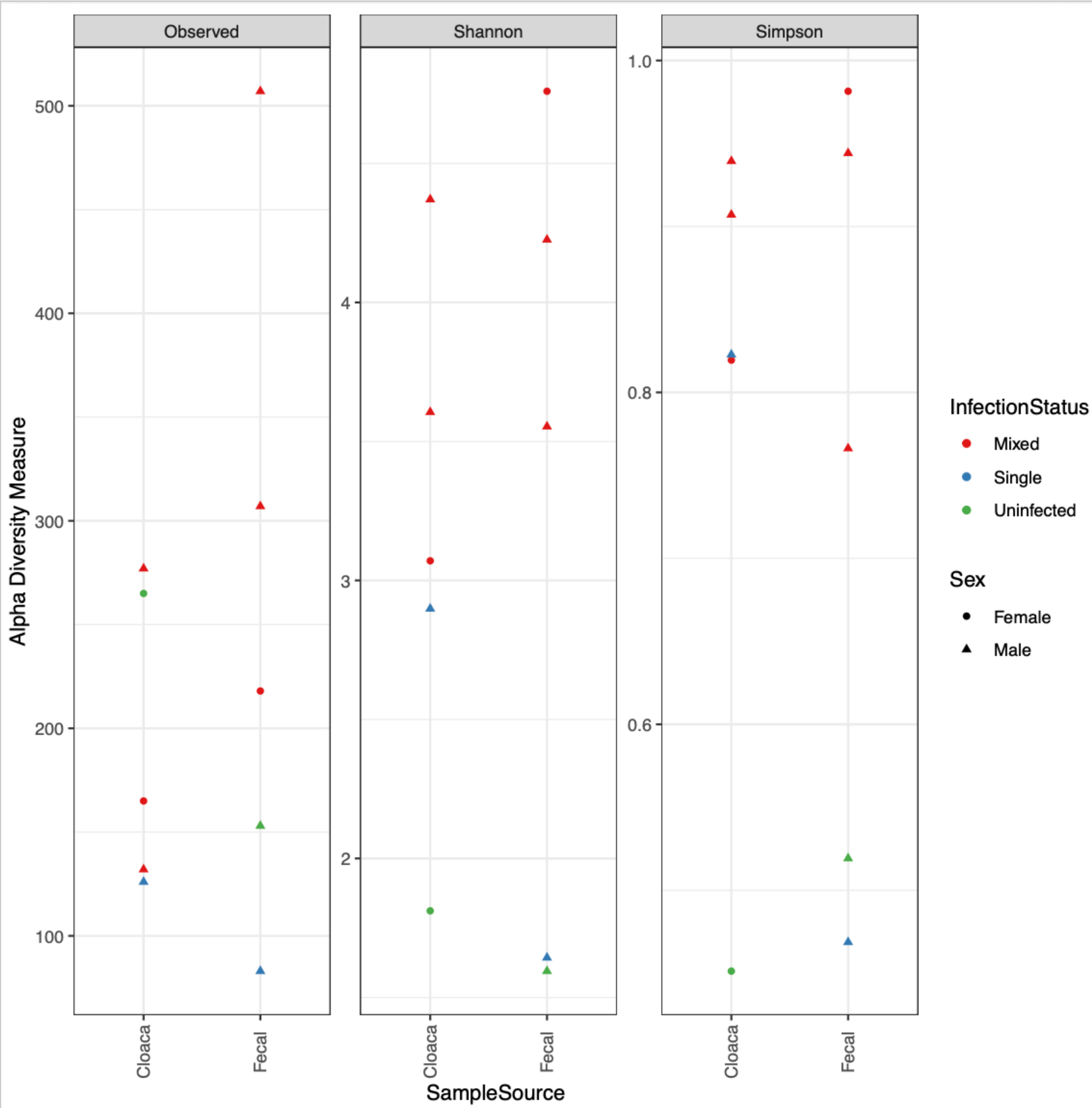
1D) Simpson diversity

What does Simpson measure?

Simpson diversity is similar to Shannon diversity in that it also considers species richness and the number of members for each species, however, it does not assign greater weight to rare species.

Upload screenshot: alpha diversity plot

Choose File ScreenShotAlphaDiversity



Grouping variable shown on the alpha diversity plot:

SampleSource and InfectionStatus

Which sample or group appears to have the greatest alpha diversity?

Of the two sample sources, fecal tends to have the greatest alpha diversity. Based on InfectionStatus, mixed has

Did Observed, Shannon, and Simpson show the same pattern?

Generally speaking, the three did show the same pattern with fecal sample source and mixed infection status having greater alpha diversity.

Part 2 — Beta diversity (NMDS with Bray-Curtis)

What beta diversity means:
Beta diversity describes differences in community composition **between samples**.

In your own words, what is beta diversity?

The diversity of organisms in one spot/sample relative to another (multiple communities being compared).

2A) Bray-Curtis dissimilarity

What does Bray-Curtis measure?

Bray-Curtis dissimilarity measures the difference in composition between two communities (0 = identical; 1 = entirely different).

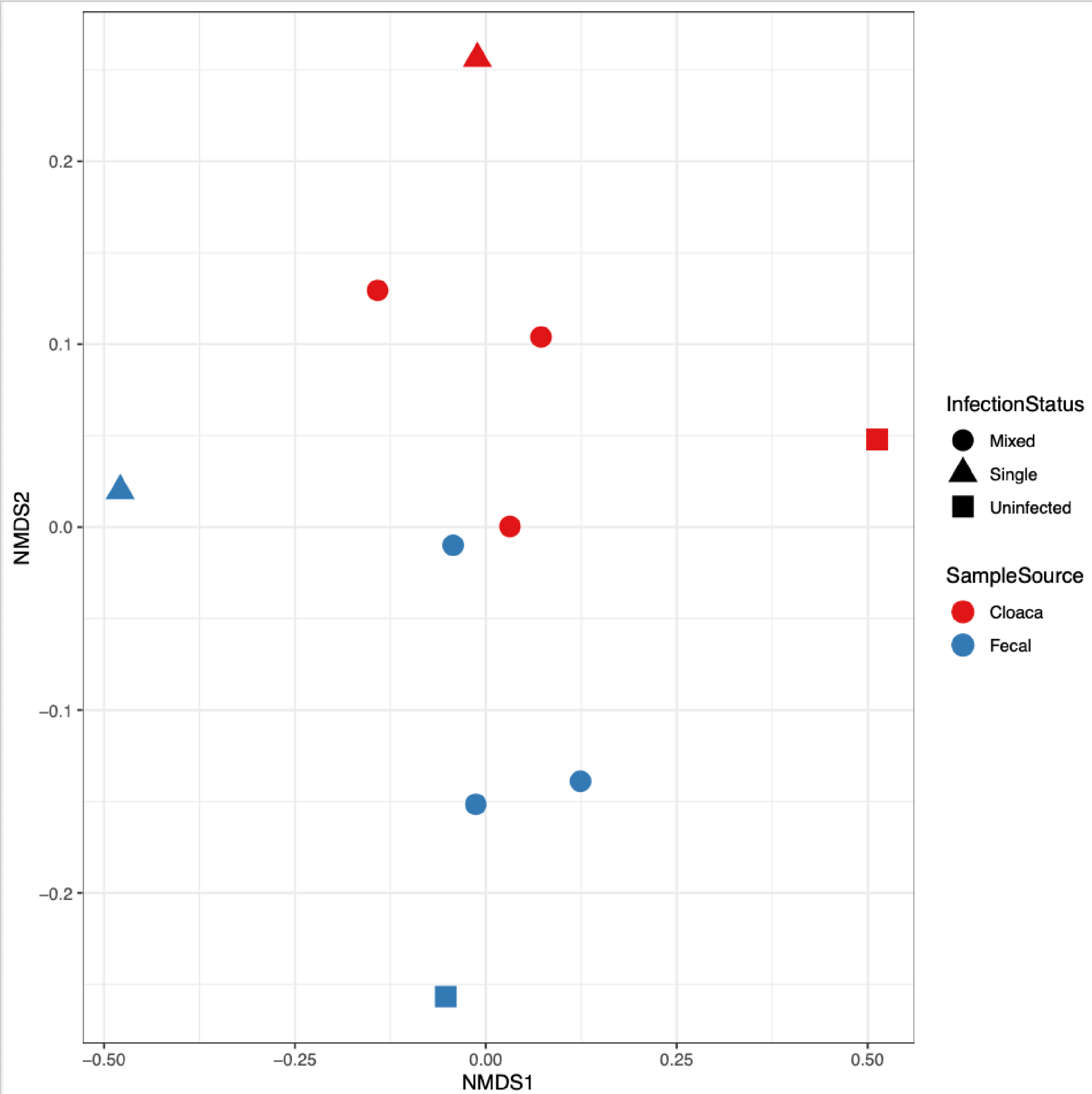
2B) NMDS

What does NMDS show?

NMDS shows a 2D plot of how similar the different samples are. The proximity of the points to one another are representative of how dissimilar the samples are (closer = greater similarity; further = less similar).

Upload screenshot: NMDS Bray-Curtis plot

Choose File ScreenShot…rayCurtisPlot



Color variable used on the NMDS plot:

SampleSource

Shape variable used on the NMDS plot (if any):

InfectionStatus

Which samples or groups cluster together?

Mixed cloaca samples tended to cluster together in the center. More broadly, samples with a mixed infection status clustered closely together in the middle and the sample sources (fecal and cloaca) were grouped with one

Are any groups clearly separated from one another? What might that suggest?

This may suggest that the microbial communities are different as samples either contained a single strain or no strain at all (largely dissimilar).

If points overlap strongly, what does that mean?

Strong overlapping of points means that the samples are very similar.

Part 3 — Taxonomic bar plot

What a taxonomic bar plot shows:
A bar plot summarizes the composition of samples by showing the relative abundance of taxa at a selected taxonomic level.

What taxonomic level did you use?

Phylum

How did you facet the plot?

By SampleSource

Why was this taxonomic level useful?

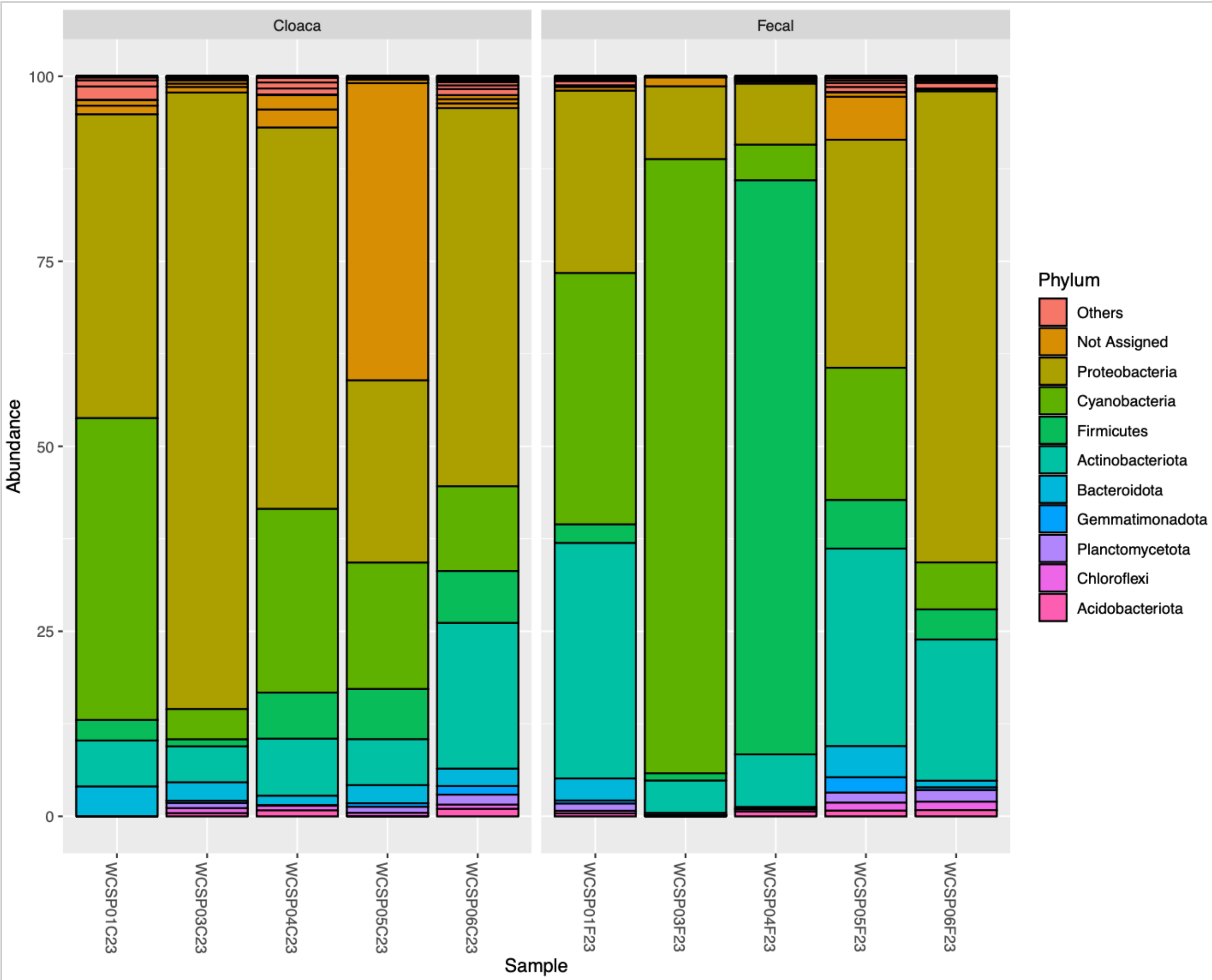
taxonomic levels were used, then there would be too little data as bacteria are categorized into more broader groups.

Why was your faceting choice useful?

based on where the sample was collected from. For instance, while Proteobacteria are the most abundant phylum from cloaca samples, Cyanobacteria seems to be increasingly abundant among fecal samples.

Upload screenshot: taxonomic bar plot

Choose File ScreenShot…rceBarChart



Which taxa appear to be most abundant?

Proteobacteria appears to be the most abundant overall.

Did taxonomic composition differ among groups?

a greater abundance of Actinobacteria among the fecal samples than in the cloca samples. Overall, the taxonomic composition does appear to slightly differ among the sample sites.

Part 4 — Overall interpretation

In 3–5 sentences, summarize what your alpha diversity, beta diversity, and bar plot together suggest about this dataset.

source (fecal or cloaca). Alpha diversity shows that fecal samples of mixed infection status showa greater amount of alpha diversity among the variables. Beta diversity resulted in the clustering of datapoints based on whether or not they were from the cloaca or fecal samples. This shows that the communities differ based on the sample site. Lastly, the bar plot showed that there were a wide array of phlya within all of the samples and the communities differ a lot based on the sample site.

Export (PDF)

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.

Submission preview

Dataset
Project/sample set: WCSMicrobiome
Grouping variable: SampleSource
Other metadata variables: BirdID, Date, InfectionStatus, HL, WBC, ParasiteLoad, Polychromasia, Sex, Control, Status, Year, Dominant, Subdominant, and Plumage.

Alpha diversity
Definition: The diversity of organisms in a given area/sample/spot.
Observed: Observed richness is a measure of the number of different species in a sample or site.
Shannon: Shannon Diversity measures how diverse a community is by considering both species richness and the number of members for each species. This measure assigns greater weight to rare species.
Simpson: Simpson diversity is similar to Shannon diversity in that it also considers species richness and the number of members for each species, however, it does not assign greater weight to rare species.
Grouping variable: SampleSource and InfectionStatus
Highest alpha diversity: Of the two sample sources, fecal tends to have the greatest alpha diversity. Based on InfectionStatus, mixed has greater alpha diversity.
Did the metrics show the same pattern? Generally speaking, the three did show the same pattern with fecal sample source and mixed infection status having greater alpha diversity.

Beta diversity (NMDS Bray-Curtis)
Definition: The diversity of organisms in one spot/sample relative to another (multiple communities being compared).
Bray-Curtis: Bray-Curtis dissimilarity measures the difference in composition between two communities (0 = identical; 1 = entirely different).
NMDS: NMDS shows a 2D plot of how similar the different samples are. The proximity of the points to one another are representative of how dissimilar the samples are (closer = greater similarity; further = less similar).
Color variable: SampleSource
Shape variable: InfectionStatus
Which groups cluster together? Mixed cloaca samples tended to cluster together in the center. More broadly, samples with a mixed infection status clustered closely together in the middle and the sample sources (fecal and cloaca) were grouped with one another.
Are any groups separated? The groups that were clearly separated were samples whose infection status were either single or uninfected. This may suggest that the microbial communities are different as samples either contained a single strain or no strain at all (largely dissimilar).
If points overlap strongly: Strong overlapping of points means that the samples are very similar.

Taxonomic bar plot
Taxonomic level: Phylum
Faceted by: By SampleSource
Why this taxonomic level? Using phylum is useful as it provides a better idea of what kind of bacteria are found in the samples which can help prevent confusion later on. If lower levels were used, there would be much more data to interpret. If higher taxonomic levels were used, then there would be too little data as bacteria are categorized into more broader groups.
Why this faceting choice? Choosing to facet the plot by infection status was useful as you can easily compare how the microbiome differs based on where the sample was collected from. For instance, while

Proteobacteria are the most abundant phylum from cloaca samples, Cyanobacteria seems to be increasingly abundant among fecal samples.
Most abundant taxa: Proteobacteria appears to be the most abundant overall.
Did composition differ among groups? In fecal samples, Cyanobacteria and Proteobacteria seem to be roughly equal in abundance, but in the cloaca samples, Proteobacteria appears as though it is the dominant phylum present. Additionally, there appears to be a greater abundance of Actinobacteria among the fecal samples than in the cloca samples. Overall, the taxonomic composition does appear to slightly differ among the sample sites.

Overall interpretation
Among all three metrics, there is support for the idea that the microbiome differs depending on the sample source (fecal or cloaca). Alpha diversity shows that fecal samples of mixed infection status showa greater amount of alpha diversity among the variables. Beta diversity resulted in the clustering of datapoints based on whether or not they were from the cloaca or fecal samples. This shows that the communities differ based on the sample site. Lastly, the bar plot showed that there were a wide array of phlya within all of the samples and the communities differ a lot based on the sample site.